
Basic Mechanical Engineering

~ ME 2001 ~

Semester: Fall 2023
Department of Electrical Engineering
National University of Computer and Emerging Sciences
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Dynamics: Planar Kinematics of Rigid Body

- ☐ ABSOLUTE MOTION ANALYSIS
- ☐ RELATIVE MOTION ANALYSIS
- ☐ PROBLEMS



Planar Kinematics of Rigid Body

- To investigate rigid-body translation and angular motion about a fixed axis.
- To study planar motion using an absolute motion analysis.
- To provide a relative motion analysis of velocity and acceleration using a translating frame of reference.
- To provide a relative-motion analysis of velocity and acceleration using a rotating frame of reference.

Basic Mech. Engg.

RIGID BODY TRANSLATION



POSITION, VELOCITY, & ACCELERATION

Relating the variables, s , v , a , & t

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$

$$a \, ds = v \, dv$$

$$a_c \, dt = dv$$

$$v = \frac{ds}{dt} = v_0 + a_c t$$

$$v \, dv = a_c \, ds$$

$$\int_{v_0}^v dv = \int_0^t a_c \, dt$$

$$\int_{s_0}^s ds = \int_0^t (v_0 + a_c t) \, dt$$

$$\int_{v_0}^v v \, dv = \int_{s_0}^s a_c \, ds$$

$$v = v_0 + a_c t$$

Constant Acceleration

$$s = s_0 + v_0 t + \frac{1}{2} a_c t^2$$

Constant Acceleration

$$v^2 = v_0^2 + 2a_c(s - s_0)$$

Constant Acceleration

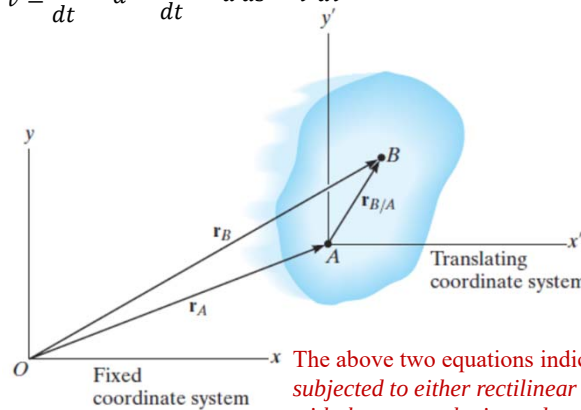
RIGID BODY TRANSLATION



POSITION, VELOCITY, & ACCELERATION

Relating the variables, s, v, a , & t

$$v = \frac{ds}{dt} \quad a = \frac{dv}{dt} \quad a ds = v dv$$



The position of B with respect to A is denoted by the *relative-position vector* $\mathbf{r}_{B/A}$ ("r of B with respect to A ").

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

Taking the first derivative,

$$\frac{d\mathbf{r}_B}{dt} = \frac{d\mathbf{r}_A}{dt} + \frac{d\mathbf{r}_{B/A}}{dt}$$

$$\mathbf{v}_B = \mathbf{v}_A + \frac{d\mathbf{r}_{B/A}}{dt}$$

$$\mathbf{v}_B = \mathbf{v}_A$$

Taking the derivative again,

$$\mathbf{a}_B = \mathbf{a}_A$$

ROTATION ABOUT A FIXED AXIS



ANGULAR DISPLACEMENT, VELOCITY, & ACCELERATION

Relating the variables, θ, ω, α , & t

$$\omega = \frac{d\theta}{dt}$$

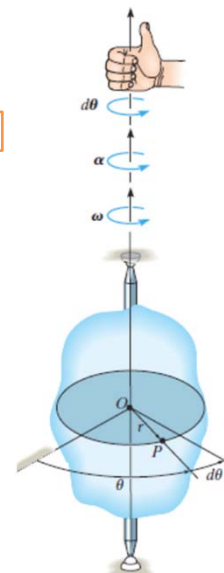
$$\alpha = \frac{d\omega}{dt}$$

$$\alpha = \frac{d^2\theta}{dt^2}$$

$$\alpha d\theta = \omega d\omega$$

$$\begin{aligned} \omega &= \omega_0 + \alpha_c t \\ \theta &= \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2 \\ \omega^2 &= \omega_0^2 + 2\alpha_c (\theta - \theta_0) \end{aligned}$$

Constant Angular Acceleration



ROTATION ABOUT A FIXED AXIS



ANGULAR DISPLACEMENT, VELOCITY, & ACCELERATION

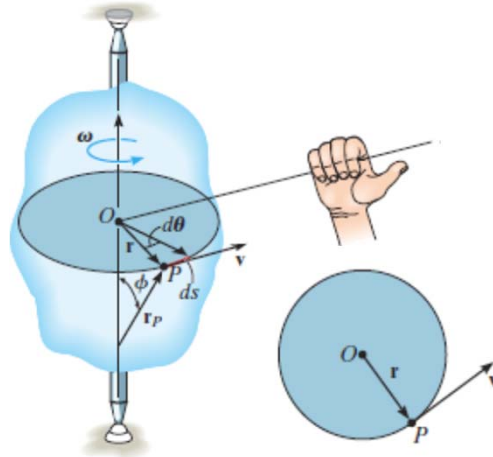
Relating the variables, θ , ω , α , & t

$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\alpha d\theta = \omega d\omega$$

$$\alpha = \frac{d^2\theta}{dt^2}$$



$$ds = r d\theta$$

$$\frac{ds}{dt} = v = r \frac{d\theta}{dt} = r\omega$$

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$$

ROTATION ABOUT A FIXED AXIS



ANGULAR DISPLACEMENT, VELOCITY, & ACCELERATION

Relating the variables, θ , ω , α , & t

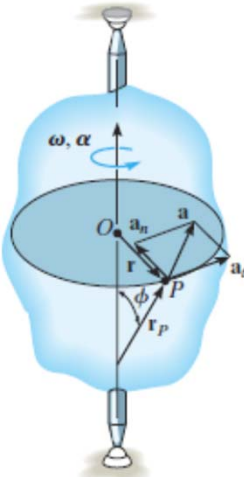
$$\omega = \frac{d\theta}{dt}$$

$$\alpha = \frac{d\omega}{dt}$$

$$\alpha d\theta = \omega d\omega$$

$$\alpha = \frac{d^2\theta}{dt^2}$$

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$$



Taking first derivative of

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\boldsymbol{\omega} \times \mathbf{r}_P)$$

$$\mathbf{a} = \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r}_P + \boldsymbol{\omega} \times \frac{d\mathbf{r}_P}{dt}$$

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r}_P + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_P)$$

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} - \omega^2 \mathbf{r}$$

$$\mathbf{a} = \mathbf{a}_t + \mathbf{a}_n$$

$$\mathbf{a}_t = \boldsymbol{\alpha} \times \mathbf{r}$$

$$\mathbf{a}_n = \omega^2 \mathbf{r}$$



Procedure for Analysis

ANGULAR MOTION

- Establish the positive sense of rotation about the axis of rotation and show it alongside each kinematic equation as it is applied.
- If a relation is known between any two of the four variables α , ω , θ and t , then a third variable can be obtained by using one of the following kinematic equations which relates all three variables

$$\omega = \frac{d\theta}{dt}, \quad \alpha = \frac{d\omega}{dt}, \quad \alpha d\theta = \omega d\omega$$

- If the body's angular acceleration is constant, then the following equations can be used:

$$\omega = \omega_0 + \alpha_c t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha_c t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$



Procedure for Analysis

MOTION OF POINT P

- In most cases the velocity of P and its two components of acceleration can be determined from the scalar equations

$$v = \omega r, \quad a_t = \alpha r, \quad a_n = \omega^2 r$$

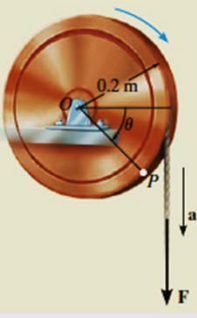
- If the geometry of the problem is difficult to visualize, the following vector equations should be used:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}_P = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{a}_t = \boldsymbol{\alpha} \times \mathbf{r}_P$$

$$\mathbf{a}_n = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_P) = -\omega^2 \mathbf{r}$$

- Here $\mathbf{r}_P$ is directed from any point on the axis of rotation to point P , whereas $\mathbf{r}$ lies in the plane of motion of P .



A cord is wrapped around a wheel in Fig. 16–5, which is initially at rest when $\theta = 0$. If a force is applied to the cord and gives it an acceleration $a = (4t) \text{ m/s}^2$, where t is in seconds, determine, as a function of time, (a) the angular velocity of the wheel, and (b) the angular position of line OP in radians.

$$(a_P)_t = \alpha r$$

$$(4t) \text{ m/s}^2 = \alpha(0.2 \text{ m})$$

$$\alpha = (20t) \text{ rad/s}^2 \curvearrowright$$

$$\alpha = \frac{d\omega}{dt} = (20t) \text{ rad/s}^2$$

$$\int_0^\omega d\omega = \int_0^t 20t \, dt$$

$$\omega = 10t^2 \text{ rad/s} \curvearrowright$$

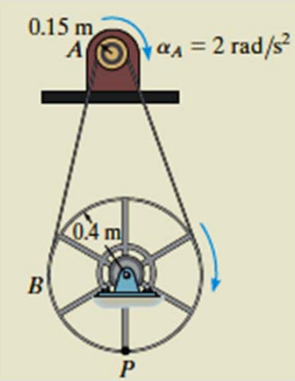
$$\frac{d\theta}{dt} = \omega = (10t^2) \text{ rad/s}$$

$$\int_0^\theta d\theta = \int_0^t 10t^2 \, dt$$

$$\theta = 3.33t^3 \text{ rad}$$

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The motor shown in the photo is used to turn a wheel and attached blower contained within the housing. The details of the design are shown in Fig. 16–6a. If the pulley A connected to the motor begins to rotate from rest with a constant angular acceleration of $\alpha_A = 2 \text{ rad/s}^2$, determine the magnitudes of the velocity and acceleration of point P on the wheel, after the pulley has turned two revolutions. Assume the transmission belt does not slip on the pulley and wheel.



$$\theta_A = 2 \text{ rev} \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 12.57 \text{ rad}$$

$$\omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

$$\omega_A^2 = 0 + 2(2 \text{ rad/s}^2)(12.57 \text{ rad} - 0)$$

$$\omega_A = 7.090 \text{ rad/s}$$

$$v = \omega_A r_A = \omega_B r_B$$

$$7.090 \text{ rad/s} (0.15 \text{ m}) = \omega_B (0.4 \text{ m})$$

$$\omega_B = 2.659 \text{ rad/s}$$

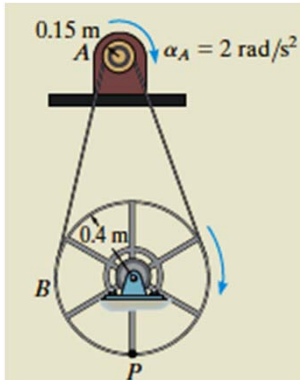
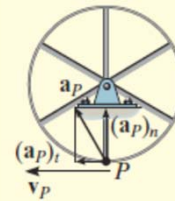
$$a_t = \alpha_A r_A = \alpha_B r_B$$

$$2 \text{ rad/s}^2 (0.15 \text{ m}) = \alpha_B (0.4 \text{ m})$$

$$\alpha_B = 0.750 \text{ rad/s}^2$$

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Motion of P .

$$v_P = \omega_B r_B = 2.659 \text{ rad/s} (0.4 \text{ m}) = 1.06 \text{ m/s}$$

$$(a_P)_t = \alpha_B r_B = 0.750 \text{ rad/s}^2 (0.4 \text{ m}) = 0.3 \text{ m/s}^2$$

$$(a_P)_n = \omega_B^2 r_B = (2.659 \text{ rad/s})^2 (0.4 \text{ m}) = 2.827 \text{ m/s}^2$$

$$a_P = \sqrt{(0.3 \text{ m/s}^2)^2 + (2.827 \text{ m/s}^2)^2} = 2.84 \text{ m/s}^2$$

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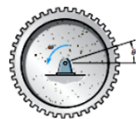
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322 CHAPTER 16 PLANAR KINEMATICS OF A RIGID BODY

FUNDAMENTAL PROBLEMS

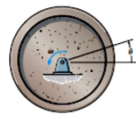
F16-1. When the gear rotates 20 revolutions, it achieves an angular velocity of $\omega = 30 \text{ rad/s}$, starting from rest. Determine its constant angular acceleration and the time required.



F16-1

16

F16-2. The flywheel rotates with an angular velocity of $\omega = (0.005\theta^2) \text{ rad/s}$, where θ is in radians. Determine the angular acceleration when it has rotated 20 revolutions.



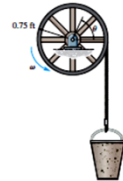
F16-2

F16-3. The flywheel rotates with an angular velocity of $\omega = (4\theta^{3/2}) \text{ rad/s}$, where θ is in radians. Determine the time it takes to achieve an angular velocity of $\omega = 150 \text{ rad/s}$. When $t = 0$, $\theta = 0$.



F16-3

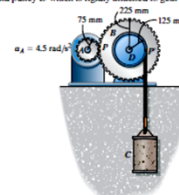
F16-4. The bucket is hoisted by the rope that wraps around a drum wheel. If the angular displacement of the wheel is $\theta = (0.5t^3 + 15t) \text{ rad}$, where t is in seconds, determine the velocity and acceleration of the bucket when $t = 3 \text{ s}$.



F16-4

F16-5. A wheel has an angular acceleration of $\alpha = (0.5\theta) \text{ rad/s}^2$, where θ is in radians. Determine the magnitude of the velocity and acceleration of a point P located on its rim after the wheel has rotated 2 revolutions. The wheel has a radius of 0.2 m and starts from rest.

F16-6. For a short period of time, the motor turns gear A with a constant angular acceleration of $\alpha_A = 4.5 \text{ rad/s}^2$, starting from rest. Determine the velocity of the cylinder and the distance it travels in three seconds. The cord is wrapped around pulley D which is rigidly attached to gear B .



F16-6

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Absolute Motion

The end of rod R shown in Fig. 16-7 maintains contact with the cam by means of a spring. If the cam rotates about an axis passing through point O with an angular acceleration α and angular velocity ω , determine the velocity and acceleration of the rod when the cam is in the arbitrary position θ .

Position Coordinate Equation.

$$x = 2r \cos \theta$$

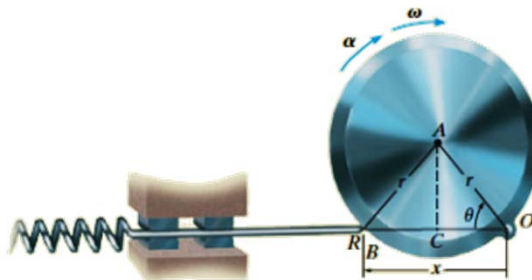
Time Derivatives.

$$\frac{dx}{dt} = -2r(\sin \theta) \frac{d\theta}{dt}$$

$$v = -2r\omega \sin \theta$$

$$\frac{dv}{dt} = -2r \left(\frac{d\omega}{dt} \right) \sin \theta - 2r\omega(\cos \theta) \frac{d\theta}{dt}$$

$$a = -2r(\alpha \sin \theta + \omega^2 \cos \theta)$$



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At a given instant, the cylinder of radius r , shown in Fig. 16-8, has an angular velocity ω and angular acceleration α . Determine the velocity and acceleration of its center G if the cylinder rolls without slipping.

Position Coordinate Equation.

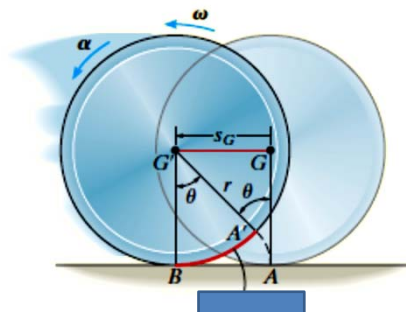
$$s_G = r\theta$$

Time Derivatives.

$$s_G = r\theta$$

$$v_G = r\omega$$

$$a_G = r\alpha$$

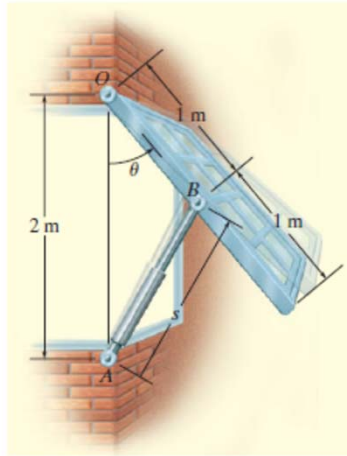


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The large window in Fig. 16-9 is opened using a hydraulic cylinder AB . If the cylinder extends at a constant rate of 0.5 m/s , determine the angular velocity and angular acceleration of the window at the instant $\theta = 30^\circ$.



Position Coordinate Equation.

$$s^2 = (2 \text{ m})^2 + (1 \text{ m})^2 - 2(2 \text{ m})(1 \text{ m}) \cos \theta$$

$$s^2 = 5 - 4 \cos \theta$$

$$\text{When } \theta = 30^\circ, \quad s = 1.239 \text{ m}$$

Time Derivatives.

$$2s \frac{ds}{dt} = 0 - 4(-\sin \theta) \frac{d\theta}{dt}$$

$$s(v_s) = 2(\sin \theta)\omega$$

$$\text{Since } v_s = 0.5 \text{ m/s, then at } \theta = 30^\circ,$$

$$(1.239 \text{ m})(0.5 \text{ m/s}) = 2 \sin 30^\circ \omega$$

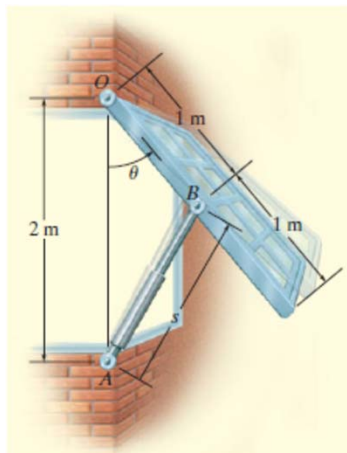
$$\omega = 0.6197 \text{ rad/s} = 0.620 \text{ rad/s}$$

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Time Derivatives.

$$s(v_s) = 2(\sin \theta)\omega$$

$$\frac{ds}{dt} v_s + s \frac{dv_s}{dt} = 2(\cos \theta) \frac{d\theta}{dt} \omega + 2(\sin \theta) \frac{d\omega}{dt}$$

$$v_s^2 + s a_s = 2(\cos \theta) \omega^2 + 2(\sin \theta) \alpha$$

$$\text{Since } a_s = dv_s/dt = 0, \text{ then}$$

$$(0.5 \text{ m/s})^2 + 0 = 2 \cos 30^\circ (0.6197 \text{ rad/s})^2 + 2 \sin 30^\circ \alpha$$

$$\alpha = -0.415 \text{ rad/s}^2$$

angular deceleration

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Relative Motion Analysis: Velocity

POSITION, DISPLACEMENT, & VELOCITY

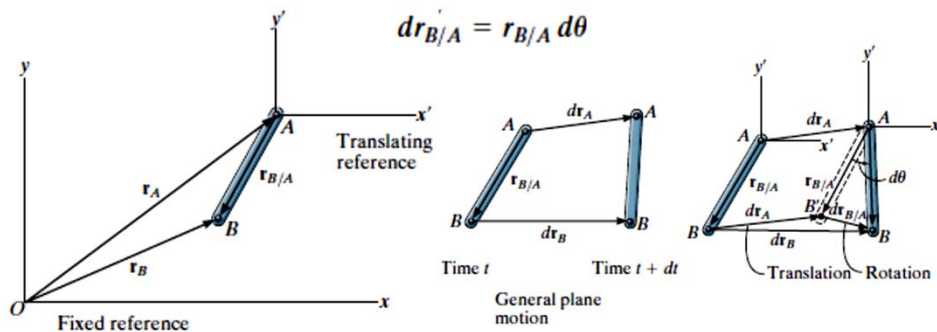
- For Position of B ,

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}$$

- During an instant of time dt , points A and B undergo displacements $d\mathbf{r}_A$ and $d\mathbf{r}_B$

$$d\mathbf{r}_B = d\mathbf{r}_A + d\mathbf{r}_{B/A}$$

due to translation of A due to rotation about A
 due to translation and rotation



Relative Motion Analysis: VELOCITY

POSITION, DISPLACEMENT, & VELOCITY

- Relationship between velocities of points A and B , taking the first derivative

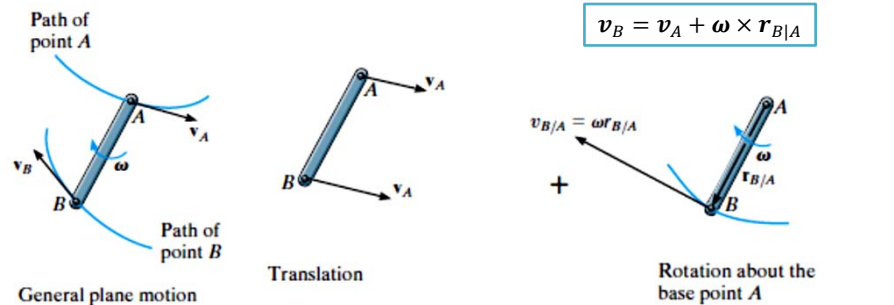
$$\frac{d\mathbf{r}_B}{dt} = \frac{d\mathbf{r}_A}{dt} + \frac{d\mathbf{r}_{B/A}}{dt}$$

Third term is $\mathbf{v}_{B/A}$, first find the magnitude,

$$\frac{dr_{B/A}}{dt} = r_{B/A} \frac{d\theta}{dt} = r_{B/A} \omega = v_{B/A}$$

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$





Procedure for Analysis: Relative Velocity

VECTOR ANALYSIS

- **Kinematic Diagram:**

- Establish the directions of the fixed x, y coordinates and draw a kinematic diagram of the body. Indicate on it the velocities of points A and B, ω and $r_{B|A}$.
- If the magnitudes of v_A , v_B , or ω are unknown, the sense or direction of these vectors can be assumed.

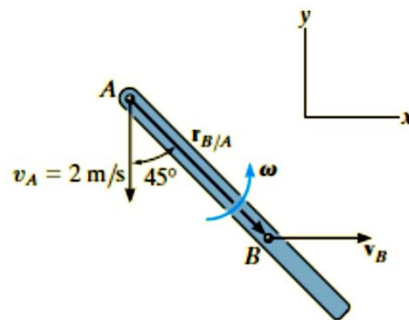
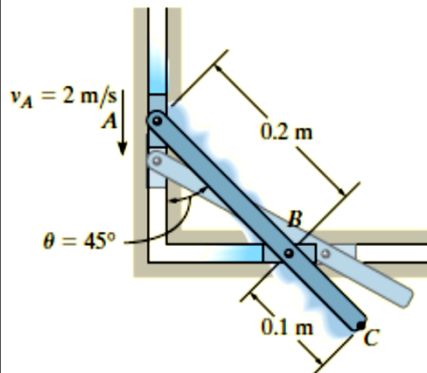
- **Velocity Equation:**

- Represent all the vectors in Cartesian vector notation,

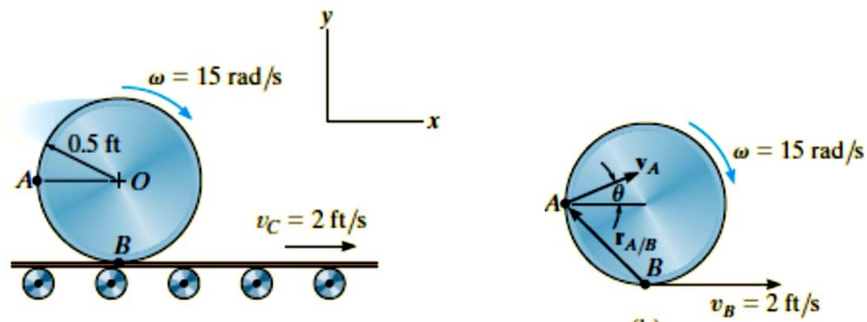
$$v_B = v_A + \omega \times r_{B|A}$$

- If the answer is *negative*, it indicates the sense of direction of the vector is opposite to that shown in kinematic diagram.

The link shown in Fig. 16–13a is guided by two blocks at A and B, which move in the fixed slots. If the velocity of A is 2 m/s downward, determine the velocity of B at the instant $\theta = 45^\circ$.



The cylinder shown in Fig. 16–14a rolls without slipping on the surface of a conveyor belt which is moving at 2 ft/s. Determine the velocity of point A. The cylinder has a clockwise angular velocity $\omega = 15 \text{ rad/s}$ at the instant shown.

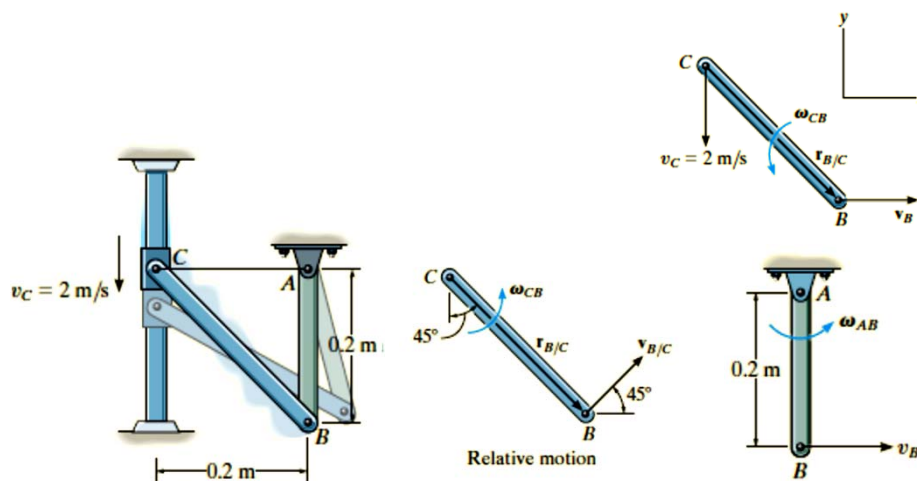


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The collar C in Fig. 16–15a is moving downward with a velocity of 2 m/s. Determine the angular velocity of CB at this instant.

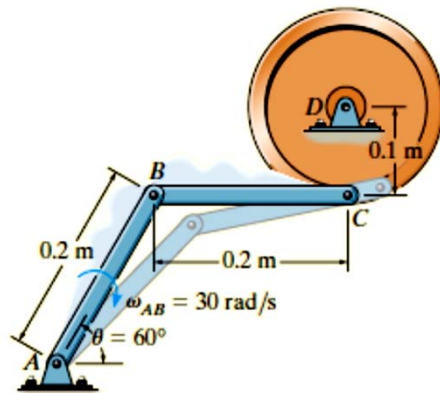


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The bar AB of the linkage shown in Fig. 16-16a has a clockwise angular velocity of 30 rad/s when $\theta = 60^\circ$. Determine the angular velocities of member BC and the wheel at this instant.

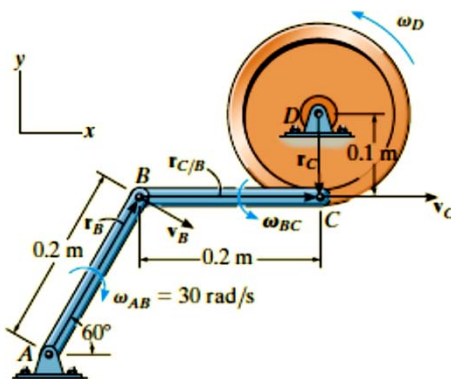


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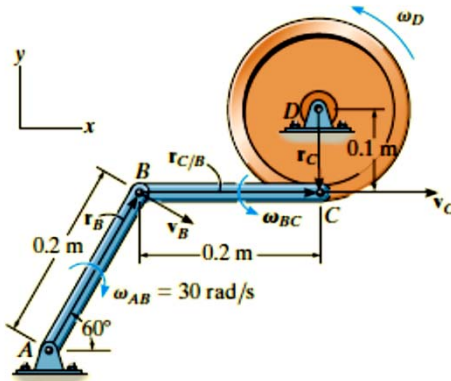


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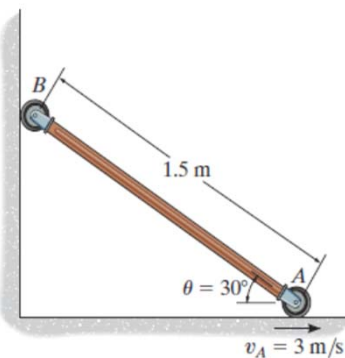


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F16–7. If roller A moves to the right with a constant velocity of $v_A = 3 \text{ m/s}$, determine the angular velocity of the link and the velocity of roller B at the instant $\theta = 30^\circ$.

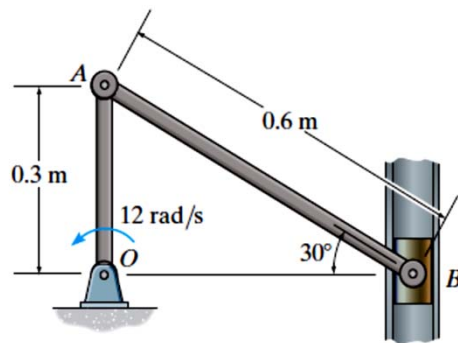


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F16-10. If crank OA rotates with an angular velocity of $\omega = 12 \text{ rad/s}$, determine the velocity of piston B and the angular velocity of rod AB at the instant shown.

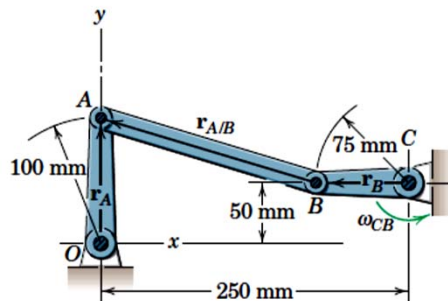


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Crank CB oscillates about C through a limited arc, causing crank OA to oscillate about O . When the linkage passes the position shown with CB horizontal and OA vertical, the angular velocity of CB is 2 rad/s counterclockwise. For this instant, determine the angular velocities of OA and AB .



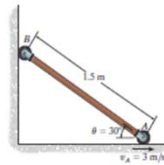
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ME101 (Spring 2017)

FUNDAMENTAL PROBLEMS

F16-7. If roller A moves to the right with a constant velocity of $v_A = 3 \text{ m/s}$, determine the angular velocity of the link and the velocity of roller B at the instant $\theta = 30^\circ$.



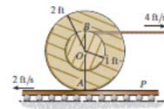
F16-7

F16-8. The wheel rolls without slipping with an angular velocity of $\omega = 10 \text{ rad/s}$. Determine the magnitude of the velocity of point B at the instant shown.



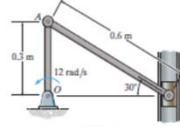
F16-8

F16-9. Determine the angular velocity of the spool. The cable wraps around the inner core, and the spool does not slip on the platform P .



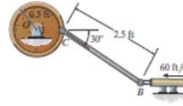
F16-9

F16-10. If crank OA rotates with an angular velocity of $\omega = 12 \text{ rad/s}$, determine the velocity of piston B and the angular velocity of rod AB at the instant shown.



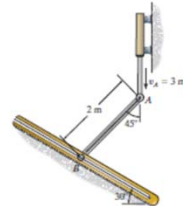
F16-10

F16-11. If rod AB slides along the horizontal slot with a velocity of 60 ft/s , determine the angular velocity of link BC at the instant shown.



F16-11

F16-12. End A of the link has a velocity of $v_A = 3 \text{ m/s}$. Determine the velocity of the peg at B at this instant. The peg is constrained to move along the slot.



F16-12

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Basic Mech. Engg.

Relative Motion Analysis: Acceleration

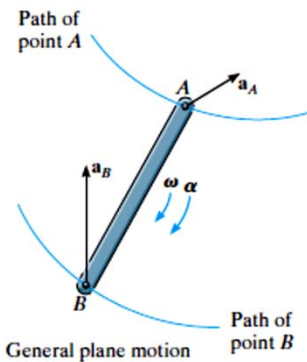


POSITION, DISPLACEMENT, & VELOCITY

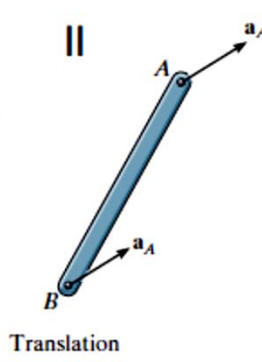
- Equation that relates acceleration of two points on a rigid bar,

$$\frac{dv_B}{dt} = \frac{dv_A}{dt} + \frac{dv_{B|A}}{dt} \text{ becomes } \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B|A}$$

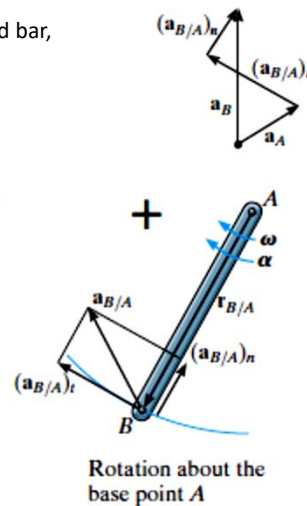
$$\mathbf{a}_B = \mathbf{a}_A + (\mathbf{a}_{B|A})_t + (\mathbf{a}_{B|A})_n$$



General plane motion



Translation



Rotation about the base point A

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ME101 (Spring 2017)

Relative Motion Analysis: Acceleration



POSITION, DISPLACEMENT, & VELOCITY

- Equation that relates acceleration of two points on a rigid bar,

$$\frac{dv_B}{dt} = \frac{dv_A}{dt} + \frac{dv_{B|A}}{dt} \text{ becomes } \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B|A}$$

$$\mathbf{a}_B = \mathbf{a}_A + (\mathbf{a}_{B|A})_t + (\mathbf{a}_{B|A})_n$$

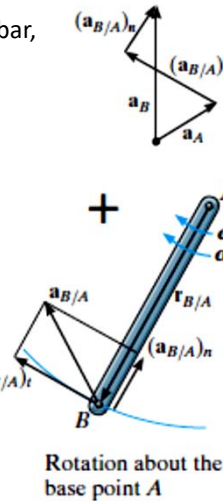
$\mathbf{a}_B$ = acceleration of point B

$\mathbf{a}_A$ = acceleration of point A

$(\mathbf{a}_{B|A})_t$ = tangential acceleration component of B with respect to A . The *magnitude* is $(\mathbf{a}_{B|A})_t = \alpha r_{B|A}$, and the *direction* is perpendicular to $\mathbf{r}_{B|A}$.

$(\mathbf{a}_{B|A})_n$ = normal acceleration component of B with respect to A . The *magnitude* is $(\mathbf{a}_{B|A})_n = \omega^2 r_{B|A}$, and the *direction* is always from B towards A .

$$\mathbf{a}_B = \mathbf{a}_A + \alpha \times \mathbf{r}_{B|A} - \omega^2 \mathbf{r}_{B|A}$$



Rotation about the base point A

Procedure for Analysis: Relative Acceleration

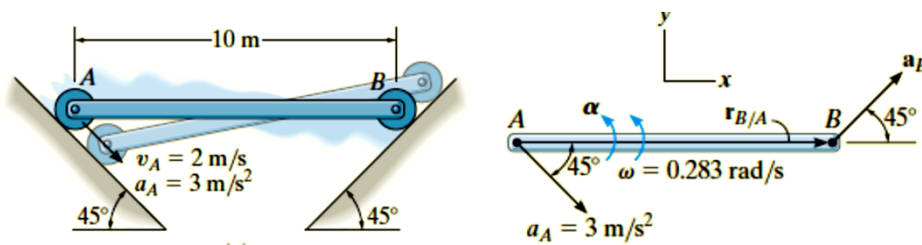


VELOCITY & VECTOR ANALYSIS

- Velocity Analysis:**
- Determine the angular velocity ω of the body using velocity analysis as already done in the previous section. Also determine the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ of points A and B if these points move along curved paths.
- Kinematic Diagram:**
- Establish the directions of the fixed x, y coordinates and draw a kinematic diagram of the body. Indicate on it $\mathbf{a}_A$, $\mathbf{a}_B$, ω , α and $\mathbf{r}_{B|A}$.
- If points A and B move along curved paths, then their accelerations should be indicated in terms of their tangential and normal components, i.e.,
- $\mathbf{a}_A = (\mathbf{a}_A)_t + (\mathbf{a}_A)_n$ and $\mathbf{a}_B = (\mathbf{a}_B)_t + (\mathbf{a}_B)_n$
- Acceleration Equation:**
- Represent all the vectors in Cartesian vector notation,

$$\mathbf{a}_B = \mathbf{a}_A + \alpha \times \mathbf{r}_{B|A} - \omega^2 \mathbf{r}_{B|A}$$

The rod AB shown in Fig. 16–27a is confined to move along the inclined planes at A and B . If point A has an acceleration of 3 m/s^2 and a velocity of 2 m/s , both directed down the plane at the instant the rod is horizontal, determine the angular acceleration of the rod at this instant.



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At a given instant, the cylinder of radius r , shown in Fig. 16–28a, has an angular velocity ω and angular acceleration α . Determine the velocity and acceleration of its center G and the acceleration of the contact point at A if it rolls without slipping.

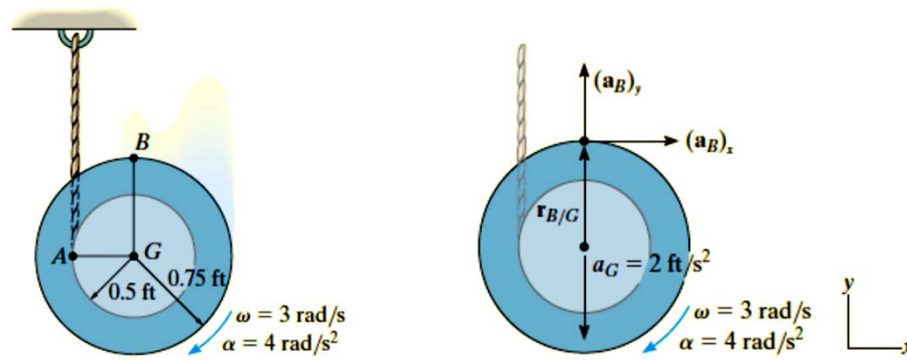


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The spool shown in Fig. 16–29a unravels from the cord, such that at the instant shown it has an angular velocity of 3 rad/s and an angular acceleration of 4 rad/s². Determine the acceleration of point B.



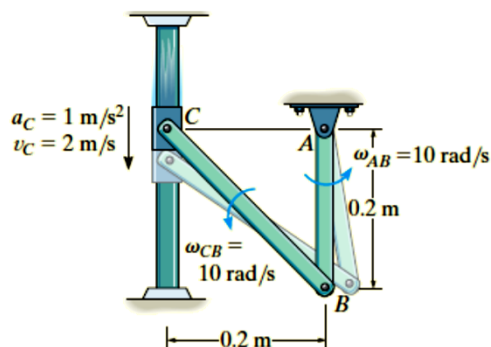
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The collar C in Fig. 16–30a moves downward with an acceleration of 1 m/s². At the instant shown, it has a speed of 2 m/s which gives links CB and AB an angular velocity $\omega_{AB} = \omega_{CB} = 10$ rad/s. (See Example 16.8.) Determine the angular accelerations of CB and AB at this instant.

$$\mathbf{v}_B = \mathbf{v}_C + \boldsymbol{\omega}_{CB} \times \mathbf{r}_{B/C}$$

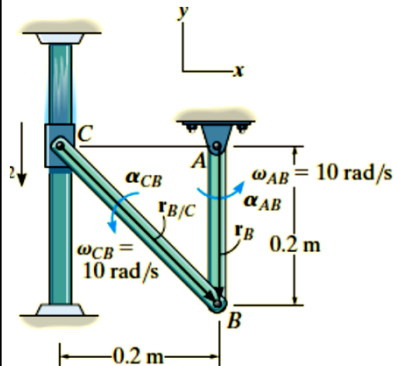


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The collar C in Fig. 16–30a moves downward with an acceleration of 1 m/s^2 . At the instant shown, it has a speed of 2 m/s which gives links CB and AB an angular velocity $\omega_{AB} = \omega_{CB} = 10 \text{ rad/s}$. (See Example 16.8.) Determine the angular accelerations of CB and AB at this instant.

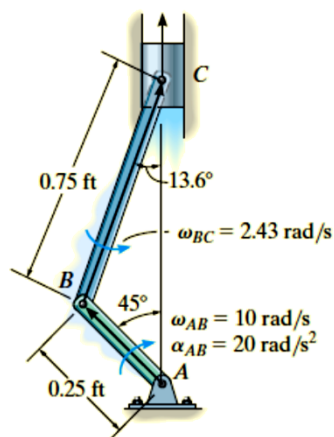


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The crankshaft AB turns with a clockwise angular acceleration of 20 rad/s^2 , Fig. 16–31a. Determine the acceleration of the piston at the instant AB is in the position shown. At this instant $\omega_{AB} = 10 \text{ rad/s}$ and $\omega_{BC} = 2.43 \text{ rad/s}$ (See Example 16.13.)

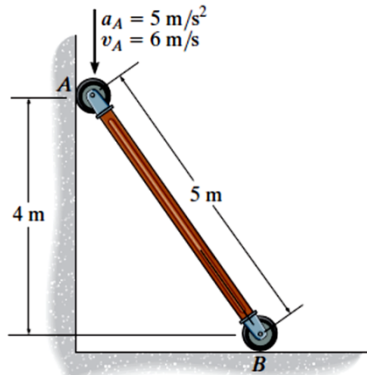


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F16-19. At the instant shown, end A of the rod has the velocity and acceleration shown. Determine the angular acceleration of the rod and acceleration of end B of the rod.

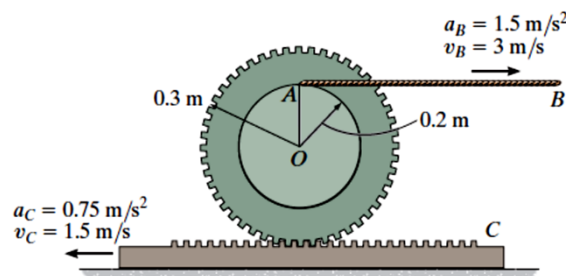


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F16-22. At the instant shown, cable AB has a velocity of 3 m/s and acceleration of 1.5 m/s^2 , while the gear rack has a velocity of 1.5 m/s and acceleration of 0.75 m/s^2 . Determine the angular acceleration of the gear at this instant. (α)



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